

SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 06: Simple Linear Regression and Multiple Linear Regression

LECTURE W6L1: Simple Linear Regression

BY

Prof. Rajib Maity

Professor, Department of Civil Engineering
Chairperson, School of Water Resources
Indian Institute of Technology Kharagpur
Kharagpur – 721302, West Bengal

CONCEPTS COVERED

Simple Linear Regression: Concept and Example

NPTEL

Introduction to Regression

- Many applications in engineering and machine learning deal with studying the relationship between the associated variables.
- The target variable is known as dependent variable, whereas other variables are known as independent variables.
- In statistics, the procedure of developing such relationship between dependent and independent variables is called regression analysis.
- The fitted statistical model is termed as regression model.
- Such models can be used for assessment of the dependent variable, knowing the independent variables.
- There are different types of regression models, and every regression model consists of some mathematical formulation with parameters to relate independent variables to dependent variable.

Simple Linear Regression

- It is one of the most commonly used models, which is based on the assumption of a linear relationship between two variables.
- In this model, the aim is to represent a dependent variable in terms of linear equation of single independent variable.
- For example, if river flow is estimated using rainfall, then, the river flow is the dependent variable Y, whereas rainfall is the independent variable X.
- It may be noted that observed values of dependent variable Y may vary even for a specific value of independent variable X owing to the uncertainty associated with it arising from unknown factors.
- Hence, Y is a random variable whose distribution is dependent on X.

Simple Linear Regression: Equation

- For random variable Y, whose distribution is dependent on X, the relationship between X and the mean of the distribution of Y is referred to as regression curve of Y on X.
- Considering the regression curve to be linear, the linear regression equation is given by:

$$Y = \alpha + \beta X + \varepsilon$$

where, ε is the difference between observed Y and the estimated Y (represented as $\hat{Y}$), termed as residual.

- The value of ε will depend on the error in measurement, and the influence of the unknown factors on Y.

Simple Linear Regression: Assumptions

For fitting a linear regression, following assumptions are made:

i. The relationship between the X and mean of Y is a straight line. Hence, for a specific value of X and Y (represented by x_i and y_i, respectively), the regression model is given by:

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

the random variables y_i are independently normally distributed with a mean of $\alpha + \beta x_i$ and variance equal to standard error (represented by σ^2), for $i = \{1, 2, \ldots, n\}$.

ii. The residuals (ε_i) are independent and normally distributed with a mean of zero and the variance of σ^2.

Simple Linear Regression: Regression Line

- The estimates of α and β (say a and b) can be calculated using observed values.
- Hence, the estimated dependent variable (denoted as $\hat{Y}$) is given by:

$$\hat{Y} = a + bX$$

Here, it should be noted that the estimated Y is the most expected value of Y given X. In other words, $\hat{Y}$ is mean of distribution of Y given X.

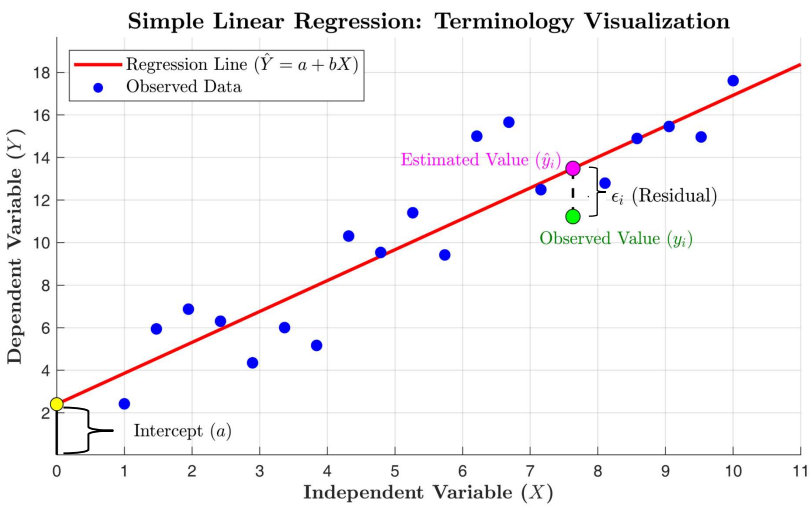
- The above equation is an equation of straight line with slope b and intercept a.
- This line is called fitted or estimated regression line.

Simple Linear Regression: Residuals

- Due to uncertainty of Y, the $\hat{Y}$ differs from Y (the difference is termed residual, as stated before).
- The ith residual (ε_i) is given by:

$$\varepsilon_i = y_i - \hat{y}_i$$ where $\hat{y}_i$ is the estimate for y_i (i.e., i^{th} observation of Y).

- The aim of regression line fitting is to get the estimate of α and β (say a and b), such that the prediction errors are minimum.
- It is not possible to minimize all the errors simultaneously, and prediction error may have positive or negative values.
- Therefore, a sign-independent criterion is needed, such as minimization of either $\sum_{i=1}^{n}|\varepsilon_i|$ or $\sum_{i=1}^{n}\varepsilon_i^2$.



Simple Linear Regression: Method of Least Squares

- Mathematically, working with absolute values is difficult as compared to working with square function.
- Hence, $\sum_{i=1}^{n} \varepsilon_i^2$ is minimized to get the estimate of α and β (i.e., a and b).
- As the sum of squared errors is minimized for estimation of regression parameters, this method is called the method of least squares.
- The estimated parameters are called least squares estimates.
- The method of least squares is described as follows:
  - The sum of squared errors can be expressed as a function of parameters as:

$$S(\alpha, \beta) = \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} [y_i - (\alpha + \beta x_i)]^2$$

Simple Linear Regression: Method of Least Squares

- If a and b are estimates of α and β such that the sum of squared errors is minimized when $\alpha = a$ and $\beta = b$, then partial derivative of S with respect to α and β at $\alpha = a$ and $\beta = b$ should be zero.

$$\left.\frac{\partial S}{\partial \alpha}\right|_{\alpha=a,\beta=b} = -2\sum_{i=1}^{n}(y_i - a - bx_i) = 0$$

Hence, $\sum_{i=1}^{n}(y_i - a - bx_i) = 0 \ldots\ldots\ldots (5)$

$$\left.\frac{\partial S}{\partial \beta}\right|_{\alpha=a,\beta=b} = -2\sum_{i=1}^{n}x_i(y_i - a - bx_i) = 0$$

Hence, $\sum_{i=1}^{n} x_i(y_i - a - bx_i) = 0$

NPTEL

Simple Linear Regression: Method of Least Squares

- Thereby, by eliminating a and solving for b, the least squares estimates of b can be written as:

$$b = [\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i/n]/[\sum_{i=1}^{n} {x_i}^2 - (\sum_{i=1}^{n} x_i)^2/n]$$

$$b = \frac{S_{xy}}{S_{xx}}$$

where, $S_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n}$ and $S_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{(\sum_{i=1}^{n} x_i)^2}{n}$

NPTEL

Simple Linear Regression: Method of Least Squares

- Finally, the least squares estimates of a can be written as:

$$a = \frac{1}{n}\left(\sum_{i=1}^{n} y_i - b\sum_{i=1}^{n} x_i\right) = \bar{Y} - b\bar{X}$$

- The individual deviations of the observations y_i from their fitted values $\hat{y}_i = a + bx_i$ are called the residuals. Thus, i^{th}residual is expressed by:

$$\varepsilon_i = y_i - a - bx_i$$

- The minimum value of the sum of square prediction errors is called residual sum of squares (RSS) or sum of squared errors (SSE).

$$SSE = \sum_{i=1}^{n} (y_i - a - bx_i)^2$$

Example – 1

An ML Engineer is optimizing the training pipeline of a CNN model and wants to estimate the training time for different dataset sizes. To study this relationship, 24 benchmark training runs were conducted. For each run, the dataset size ((X), in GB) and time required to complete one training epoch ((Y), in hours) were recorded. The observations are provided in next slide.

Assuming the training time varies linearly with the dataset size:

(a) Fit a Simple Linear Regression model to the data. Calculate the optimal weights (the slope and the intercept) by minimizing the sum of squared errors.

(b) Using the fitted model, predict the exact time required to complete one epoch for a dataset of size 14 GB.

Example – 1

Run No.	X (in GB)	Y (in hr.)	Run No.	X (in GB)	Y (in hr.)
1	6.9	5.4	13	5.5	4.4
2	6.4	4.1	14	11.4	9.3
3	6.5	4.7	15	10.8	7.4
4	5.1	3.5	16	7.5	4.8
5	7.1	4.8	17	8.2	7.2
6	7.1	5.0	18	7.9	5.9
7	10.2	7.2	19	4.1	3.0
8	9.9	6.0	20	5.0	4.3
9	8.4	4.7	21	6.7	4.3
10	5.8	6.4	22	4.3	3.0
11	10.1	7.4	23	10.4	8.9
12	7.3	5.8	24	3.9	5.4

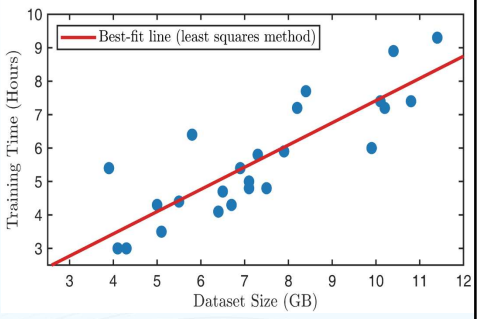
Solution:

Given:

Independent Variable, X = Training Dataset Size (in GB)

n = total no. of runs = 24

Dependent Variable, Y = Time to complete one training epoch (in hours)



(a) A scatterplot of the variables reveals a linear relationship (as shown in the figure).

Hence, the relationship between X and expected value of Y, being linear, is expressed as:

$$\hat{Y} = bX + a$$

The calculations are presented in the table (in the next slide).

Solution:

Run No	y_i	x_i	x_i^2	$x_i y_i$	y_i^2	Run No	y_i	x_i	x_i^2	$x_i y_i$	y_i^2
1	5.40	6.90	47.61	37.26	29.16	13	4.40	5.50	30.25	24.20	19.36
2	4.10	6.40	40.96	26.24	16.81	14	9.30	11.40	129.96	106.02	86.49
3	4.70	6.50	42.25	30.55	22.09	15	7.40	10.80	116.64	79.92	54.76
4	3.50	5.10	26.01	17.85	12.25	16	4.80	7.50	56.25	36.00	23.04
5	4.80	7.10	50.41	34.08	23.04	17	7.20	8.20	67.24	59.04	51.84
6	5.00	7.10	50.41	35.50	25.00	18	5.90	7.90	62.41	46.61	34.81
7	7.20	10.20	104.04	73.44	51.84	19	3.00	4.10	16.81	12.30	9.00
8	6.00	9.90	98.01	59.40	36.00	20	4.30	5.00	25.00	21.50	18.49
9	4.70	8.40	70.56	39.48	22.09	21	4.30	6.70	44.89	28.81	18.49
10	6.40	5.80	33.64	37.12	40.96	22	3.00	4.30	18.49	12.90	9.00
11	7.40	10.10	102.01	74.74	54.76	23	8.90	10.40	108.16	92.56	79.21
12	5.80	7.30	53.29	42.34	33.64	24	5.40	3.90	15.21	21.06	29.16

Totals	Values
$\sum_{i=1}^{n} y_i$	132.90
$\sum_{i=1}^{n} x_i$	176.50
$\sum_{i=1}^{n} x_i^2$	1410.51
$\sum_{i=1}^{n} x_i y_i$	1048.92
$\sum_{i=1}^{n} y_i^2$	801.29

Solution:

The parameter b can be calculated as:

$$S_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n} = 1048.92 - (176.50 \times 132.90)/24 = 71.55$$

$$S_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} = 1410.51 - (176.50)^2/24 = 112.50$$

$$b = \frac{S_{xy}}{S_{xx}} = \frac{71.55}{112.50} = 0.636$$

The parameter a is calculated as:

$$a = \frac{1}{n}\left(\sum_{i=1}^{n} y_i - b \sum_{i=1}^{n} x_i\right) = (132.9 - 0.636 \times 176.5)/24 = 0.86$$

Hence, the regression equation is : $\hat{Y} = 0.636X + 0.86$

(b) Therefore, the exact time it will take to complete one epoch on training dataset of size 14 GB is,

$$= 0.636 * 14 + 0.86 = 9.76 \text{ hrs} \approx 10 \text{ hrs}$$


FREE ONLINE EDUCATION
swayam
Thank You!



SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 06: Simple Linear Regression and Multiple Linear Regression

LECTURE W6L2: Curvilinear Regression

BY

Prof. Rajib Maity

Professor, Department of Civil Engineering
Chairperson, School of Water Resources
Indian Institute of Technology Kharagpur
Kharagpur – 721302, West Bengal

CONCEPTS COVERED

Curvilinear Regression: Concept and Example

NPTEL

Introduction to Curvilinear Regression

- In previous lecture, simple linear regression was discussed where the mean of the distribution of Y is given by $\alpha + \beta x$ for a particular value of X.
- But there are some cases where the regression curve is non-linear, but the least squares method of analysis is still applicable.
- Such cases of regression are called curvilinear or non-linear regression.

- These regression models are classified into two categories:
  i. Model transformable to linear regression.
  ii. Model not transformable to linear regression.

NPTEL

Model Transformable to Linear Regression

- Some of the curvilinear regression models, if transformed, can be converted into linear regression models.
- After transformation, the least squares estimates of parameter can be obtained by the method of least squares (as explained in the previous lecture).
- Two very commonly used relationships that can be fitted using the least squares method after transformation are as follows:

i. Reciprocal function:

$$y = \frac{1}{\alpha + \beta x}$$

It represents a linear relationship between x and $1/y$, as follows:

$$\frac{1}{y} = \alpha + \beta x$$

NPTEL

Model Transformable to Linear Regression

ii. Power Function:

$$y = \alpha x^{\beta}$$

It represents a linear relationship between $\log(x)$ and $\log(y)$, as follows:

$$\log y = \log \alpha + \beta \log x$$

Example – 1

A dataset containing an independent variable (X) and a dependent variable (Y) is provided. It is assumed that the relationship between X and Y follows a reciprocal decay pattern. Fit a reciprocal regression model to this data using the Method of Least Squares.

X	13.50	17.30	30.00	14.20	15.40	15.37	14.87	14.02	10.54	17.38	20.01
Y	9.51	12.07	15.84	12.65	8.73	14.34	11.99	11.54	10.84	10.32	12.83

X	16.57	20.50	15.53	17.48	14.04	13.70	28.37	17.74	23.24	12.86
Y	11.32	14.43	11.31	14.20	13.26	11.63	16.83	13.84	13.08	10.53

Solution:

The general form of a simple reciprocal function is given by:

$$Y = \frac{1}{\alpha + \beta X}$$

Because this equation is non-linear with respect to its parameters, standard linear regression cannot be applied directly. It must be transformed into a linear form.

By taking the reciprocal of both sides,

$$\frac{1}{Y} = \alpha + \beta X$$

Let $Y' = \frac{1}{Y}$. The equation now takes the standard linear form:

$$Y' = \alpha + \beta X$$

where, Y' is the transformed dependent variable, X remains the independent variable, α is the intercept of the transformed linear model and β is the slope of the transformed linear model.

Solution:

To use the Method of Least Squares, the summations of X, Y', X^2, and XY' need to be calculated. The table outlines the necessary calculations.

Sl. No.	Y'_i	X_i	X_i^2	$X_iY'_i$	Sl. No.	Y'_i	X_i	X_i^2	$X_iY'_i$
1	0.11	13.50	182.25	1.42	12	0.09	16.57	274.73	1.46
2	0.08	17.30	299.29	1.43	13	0.07	20.50	420.19	1.42
3	0.06	30.00	900.00	1.89	14	0.09	15.53	241.10	1.37
4	0.08	14.20	201.64	1.12	15	0.07	17.48	305.54	1.23
5	0.11	15.40	237.16	1.76	16	0.08	14.04	197.17	1.06
6	0.07	15.37	236.11	1.07	17	0.09	13.70	187.66	1.18
7	0.08	14.87	221.07	1.24	18	0.06	28.37	805.03	1.69
8	0.09	14.02	196.42	1.21	19	0.07	17.74	314.55	1.28
9	0.09	10.54	111.01	0.97	20	0.08	23.24	540.31	1.78
10	0.10	17.38	302.09	1.68	21	0.09	12.86	165.39	1.22
11	0.08	20.01	400.41	1.56					

Totals	Values
$\sum_{i=1}^{n} Y'_i$	1.73
$\sum_{i=1}^{n} X_i$	362.61
$\sum_{i=1}^{n} X_i^2$	6739.12
$\sum_{i=1}^{n} X_iY'_i$	29.07

Solution:

The parameter β can be calculated as: $\beta = \frac{S_{XY}}{S_{XX}}$

$$S_{XY} = \sum_{i=1}^{n} X_i Y_i - \frac{\sum_{i=1}^{n} X_i \sum_{i=1}^{n} Y_i}{n} = (29.07) - \big((362.61) \times 1.73\big)/21 = -0.80$$

$$S_{XX} = \sum_{i=1}^{n} X_i^2 - \frac{\left(\sum_{i=1}^{n} X_i\right)^2}{n} = 6739.12 - (362.61)^2/21 = 477.88$$

Thus, $\beta = \frac{S_{xy}}{S_{xx}} = \frac{-0.80}{477.88} = -0.00167$

The parameter α is calculated as:

$$\alpha = \frac{1}{n}\left(\sum_{i=1}^{n} Y_i - b \sum_{i=1}^{n} X_i\right) = (1.73 - (-0.00167) \times (362.61))/21 = 0.111$$

The final curvilinear regression model would be written as:

$$\frac{1}{Y} = 0.111 - 0.00167X \text{ or } Y = \frac{1}{0.111 - 0.00167X}$$

Example – 2

The maximum rate at which soil can absorb water at a given instance of time is defined as infiltration capacity (f_t). Multiple models exist for modeling infiltration rate with respect to time. In a flooding type infiltration test, infiltration capacity data are recorded. For most of soil type, the infiltration capacity at initial time is highest (known as initial infiltration capacity, denoted by f_0), which gradually decreases to steady-state infiltration capacity (also known as constant or ultimate infiltration capacity, denoted by f_c) at $t = t_c$.

Time since experiment starts (minutes)	5	10	20	30	45	60	75	90	105	120
Cumulative infiltration depth (cm)	1.30	2.50	4.30	5.75	7.40	8.75	9.90	10.95	11.90	12.85

Example – 2

a) Plot the curves for:
   i. Infiltration capacity versus time
   ii. Infiltration capacity versus cumulative infiltration
   iii. Cumulative infiltration versus time
   iv. $ln(f_t - f_c)$ versus time

b) Find the least squares estimate of the Horton's infiltration model parameter given by:

$$f_t = f_c + (f_0 - f_c)e^{-kt}$$

where, f_t is the infiltration capacity at time t. Similarly, f_0 is the infiltration capacity at $t = 0$ and f_c is the constant infiltration capacity at $t = t_c$.

c) Fit a Kostiakov Infiltration Model over the data given by:

$$F_t = at^b$$

where, F_t is the cumulative infiltration capacity at time t.

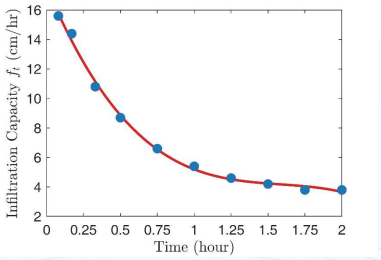
Solution:

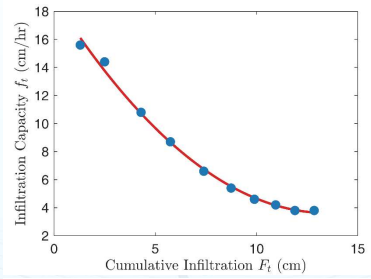
a) Incremental infiltration depths along with various other parameters are calculated and shown in the table.

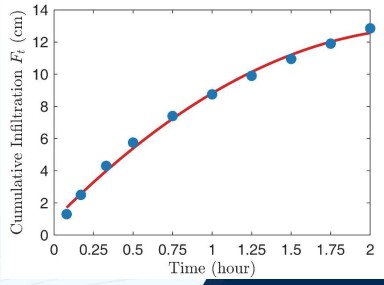
Time since experiment start (t) (minutes)	Cumulative Infiltration Depth (F_t) (cm)	Incremental Infiltration Depth ($F_t - F_{t-1}$) (cm)	t (in hr)	Infiltration Capacity f_t=($F_t - F_{t-1}$)/Δt (cm/hr)	$(\frac{1}{F_t})$	$ln(f_t - f_c)$
0						
5	1.30	1.30	0.08	15.60	0.77	2.47
10	2.50	1.20	0.17	14.40	0.40	2.36
20	4.30	1.80	0.33	10.80	0.23	1.95
30	5.75	1.45	0.50	8.70	0.17	1.59
45	7.40	1.65	0.75	6.60	0.14	1.03
60	8.75	1.35	1.00	5.40	0.11	0.47
75	9.90	1.15	1.25	4.60	0.10	-0.22
90	10.95	1.05	1.50	4.20	0.09	-0.92
105	11.90	0.95	1.75	3.80	0.08	
120	12.85	0.95	2.00	3.80	0.08	

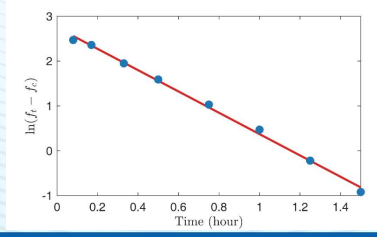
Solution:

a) The relationship between different quantities is shown graphically in following figures.









Solution:

b) Horton's infiltration equation can be transformed to linear equation as:

$f_t = f_c + (f_0 - f_c)e^{-kt} \Rightarrow f_t - f_c = (f_0 - f_c)e^{-kt} \Rightarrow \ln(f_t - f_c) = \ln(f_0 - f_c) - kt$

Hence, by comparing this form and a linear regression model with $\ln(f_t - f_c)$ as y and t as x :

$$\hat{y} = a + bx$$

From the table, $f_c = 3.8$, $f_0 = 15.6$, $a = \ln(f_0 - f_c) = \ln(15.6 - 3.8) = 2.468$ and $b = -k$.

Then,

$a = \bar{y} - b\bar{x} \Rightarrow 2.468 = \bar{y} + k\bar{x} \Rightarrow 2.468 = 1.091 + k(0.6975)$

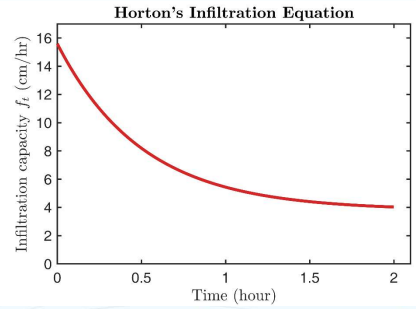
$\Rightarrow k = \frac{(2.468 - 1.091)}{0.6975} = 1.974$

Please note that $\bar{x}$ is computed using only first 8 values of x, i.e., t, since only 8 values of y, i.e., $\ln(f_t - f_c)$, are available.

Solution:

Hence, fitted Horton's equation is given by:

$$\ln(f_t - 3.8) = 2.468 - 1.974t$$
$$\Rightarrow f_t - 3.8 = e^{2.468}e^{-1.974t}$$
$$\Rightarrow f_t = 3.8 + 11.8e^{-1.974t}$$



c) The Kostiakov infiltration model can be transformed to linear equation as :

$$F_t = at^b \Rightarrow ln\,(F_t) = ln\,(a) + b\,ln\,(t)$$

$ln\,(F_t)$ and $ln\,(t)$ are shown in the table in the next slide

The estimates of parameters of this equation ($ln(a)$ and b) can be obtained by preparing another table to evaluate the required parameters where, $x_i = ln(t)$ and $y_i = ln(F_t)$.

Solution:

Time since experiment start (t) (minutes)	Cumulative Infiltration Depth (F_t) (cm)	t (hrs)	$ln\,(F_t)$	$ln\,(t)$
5	1.30	0.08	0.26	-2.48
10	2.50	0.17	0.92	-1.79
20	4.30	0.33	1.46	-1.10
30	5.75	0.50	1.75	-0.69
45	7.40	0.75	2.00	-0.29
60	8.75	1.00	2.17	0.00
75	9.90	1.25	2.29	0.22
90	10.95	1.50	2.39	0.41
105	11.90	1.75	2.48	0.56
120	12.85	2.00	2.55	0.69

Solution:

Sl. No.	Y_i	X_i	X_i^2	X_iY_i
1	0.26	-2.48	6.17	-0.65
2	0.92	-1.79	3.21	-1.64
3	1.46	-1.10	1.21	-1.60
4	1.75	-0.69	0.48	-1.21
5	2.00	-0.29	0.08	-0.58
6	2.17	0.00	0.00	0.00
7	2.29	0.22	0.05	0.51
8	2.39	0.41	0.16	0.97
9	2.48	0.56	0.31	1.39
10	2.55	0.69	0.48	1.77
Total	18.27	-4.47	12.16	-1.05

Totals	Values
$\sum_{i=1}^{n} Y_i$	18.27
$\sum_{i=1}^{n} X_i$	-4.47
$\sum_{i=1}^{n} X_i^2$	12.16
$\sum_{i=1}^{n} X_iY_i$	-1.05

Solution:

The parameter b can be calculated as: $b = \frac{S_{XY}}{S_{XX}}$

$$S_{XY} = \sum_{i=1}^{n} X_iY_i - \frac{\sum_{i=1}^{n} X_i \sum_{i=1}^{n} Y_i}{n} = (-1.05) - \big((-4.47) \times 18.27\big)/10 = 7.13$$

$$S_{XX} = \sum_{i=1}^{n} X_i^2 - \frac{\left(\sum_{i=1}^{n} X_i\right)^2}{n} = 12.16 - (-4.47)^2/10 = 10.16$$

Thus, $b = \frac{S_{XY}}{S_{XX}} = \frac{7.13}{10.16} = 0.702$

The parameter $\ln(a)$ is calculated as:

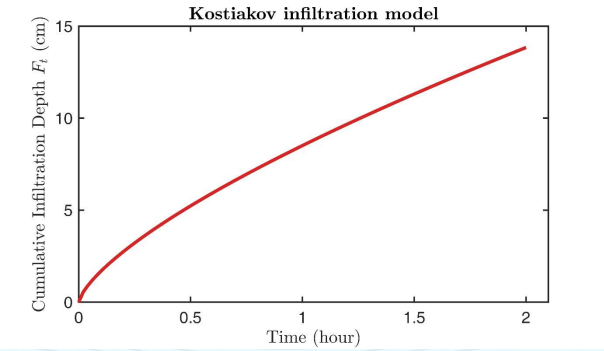
$$ln\,(a) = \frac{1}{n}\left(\sum_{i=1}^{n} Y_i - b\sum_{i=1}^{n} X_i\right) = (18.27 - 0.702 \times (-4.47))/10 = 2.141$$

$$\Rightarrow a = e^{2.141} = 8.508$$

Solution:

Therefore, Kostiakov infiltration model for the observed infiltration data is given by,

$$F_t = 8.508t^{0.702}$$



Model Not Transformable to Linear Regression

- This is the case where the functional form of the regression Y on X is not transformable to linear regression.
- For example, a polynomial fit between Y and X is given by:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_p x^p + \varepsilon \ldots\ldots\ldots (17)$$

where the degree of the equation is determined by inspecting the data.

- The corresponding coefficients can also be calculated by the method of least squares as discussed previously.
- The coefficients of fit (β_i) can be obtained by considering different powers of independent variable as separate independent variable and using the concept of multiple linear regression.


FREE ONLINE EDUCATION
swayam
Thank You!



SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 06: Simple Linear Regression and Multiple Linear Regression

LECTURE W6L3: Multiple Linear Regression

BY

Prof. Rajib Maity

Professor, Department of Civil Engineering
Chairperson, School of Water Resources
Indian Institute of Technology Kharagpur
Kharagpur – 721302, West Bengal

CONCEPTS COVERED

Multiple Linear Regression: Concept and Examples

NPTEL

Introduction to Multiple Linear Regression

- In previous lectures, the relation between a dependent and a single independent variable was discussed.
- However, in many cases, the dependent variable may depend on more than one independent variable.
  - Example: The total driving range of an electric vehicle is dependent on its battery capacity, average driving speed, external air temperature, and total vehicle payload.
- A general Multiple Linear Regression (MLR) model can be represented as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

where Y is the dependent variable and X_1, X_2,..., X_p are the independent variables; β_0, β_1, β_2,..., β_p are the unknown parameters and ε is the residual.

NPTEL

Multiple Linear Regression

- It should be noted that while fitting MLR, assumptions of simple linear regression should hold, and additionally, the data should not have multicollinearity.
- Multicollinearity represents a situation that linear combination of some inputs (independent variables) results in zero.
- Now, a set of observed data will consist of n observations of Y and corresponding n observations of p independent variables.
- Thereby, the general MLR equation can be re-written as:

$$Y_i = \beta_0 + \sum_{j=1}^{p} \beta_j X_{i,j}$$

where Y_i is the ith observation of the dependent variable and $X_{i,j}$ is the ith observation of the jth independent variable.

NPTEL

Multiple Linear Regression

- In matrix form, it can be written as :

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{1,1} & X_{1,2} & X_{1,3} & \cdots & X_{1,p} \\ 1 & X_{2,1} & X_{2,2} & X_{2,3} & \cdots & X_{2,p} \\ 1 & X_{3,1} & X_{3,2} & X_{3,3} & \cdots & X_{3,p} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n,1} & X_{n,2} & X_{n,3} & \cdots & X_{n,p} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}$$

or

$$Y = X\beta$$

where, Y is an $n \times 1$ vector of the dependent variable, X is an $n \times (p+1)$ matrix of the independent variables and β is a $(p+1) \times 1$ vector of the unknown parameters.

NPTEL

Multiple Linear Regression

- In order to find out the values of the parameters, the least squares method can be used.
- Hence, by minimizing $\sum_{i=1}^{n} \varepsilon_i^2$, $\hat{\beta}$ is obtained.
- In matrix form, the sum of squared error can be written as:

$$\sum_{i=1}^{n} \varepsilon_i^2 = e^T e = \left(Y - X\hat{\beta}\right)^T \left(Y - X\hat{\beta}\right)$$

- Differentiating the above equation with respect to $\hat{\beta}$ and setting the value of the expression to zero, following equation is obtained :

$$X^T Y = X^T X \hat{\beta}$$

- The solution for $\hat{\beta}$ can be obtained by multiplying both sides of the equation with $(X^T X)^{-1}$.
- Thus, $\hat{\beta} = (X^T X)^{-1} X^T Y$

NPTEL

Example – 1

Given a dataset consisting of three independent variables (X_1, X_2 and X_3) and one dependent variable (Y), develop a Multiple Linear Regression (MLR) model to estimate Y using the provided independent variables.

Sl.No.	Y	X_1	X_2	X_3
1	32.4	12	4	100
2	114.5	45	8	420
3	62.6	23	2	210
4	89.7	33	5	380
5	41.5	18	9	110
6	90.0	29	1	490
7	99.7	41	6	260
8	51.5	15	3	340
9	89.1	38	7	190
10	70.1	22	5	450

Solution:

The transpose of independent variable matrix (X) is given by:

$$X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 12 & 45 & 23 & 33 & 18 & 29 & 41 & 15 & 38 & 22 \\ 4 & 8 & 2 & 5 & 9 & 1 & 6 & 3 & 7 & 5 \\ 100 & 420 & 210 & 380 & 110 & 490 & 260 & 340 & 190 & 450 \end{bmatrix}^T$$

Similarly, the transpose of dependent variable matrix (Y) is given by:

$$Y = [32.4 \quad 114.5 \quad 62.6 \quad 89.7 \quad 41.5 \quad 90.0 \quad 99.7 \quad 51.5 \quad 89.1 \quad 70.1]^T$$

Now, $(X^T X) = \begin{bmatrix} 10 & 276 & 50 & 2950 \\ 276 & 8786 & 1477 & 86540 \\ 50 & 1477 & 310 & 13720 \\ 2950 & 86540 & 13720 & 1048900 \end{bmatrix}$

Solution:

Also, $(X^T X)^{-1} = \begin{bmatrix} 1.4342 & -0.0048 & -0.1128 & -0.0022 \\ -0.0048 & 0.0014 & -0.0032 & -0.0001 \\ -0.1128 & -0.0032 & 0.0260 & 0.0002 \\ -0.0022 & -0.0001 & 0.0002 & 0 \end{bmatrix}$

Then, $(X^T Y) = \begin{bmatrix} 740 \\ 23090 \\ 3810 \\ 239130 \end{bmatrix}$

Therefore, $\hat{\beta} = (X^T X)^{-1} X^T Y = \begin{bmatrix} 5.40 \\ 2.10 \\ -0.80 \\ 0.05 \end{bmatrix}$

Hence, the MLR model to estimate Y using the given independent variables is obtained as:

$$E(Y|X_1, X_2, X_3) = 5.40 + 2.10X_1 - 0.80X_2 + 0.05X_3$$

Example – 2

An AI engineer is studying the factors affecting server response time. Data collected over ten observation periods include three variables (shown in the table): System Latency (the time it takes for the server to process and return a response, measured in milliseconds), Traffic Volume (the number of simultaneous user requests hitting the server) and Server Temperature (the operating temperature of the server's processing chip, measured in $°C$). It is assumed that system latency (Y) can be estimated from traffic volume (X_1) and server temperature (X_2) using Multiple Linear Regression model as given by:

$$E(Y|X_1, X_2) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

Determine the coefficients β_0, β_1, and β_2 respectively.

Example – 2

Observation No.	System Latency (Y, ms)	Traffic Volume (X_1)	Server Temperature (X_2, °C)
1	7	12	22.30
2	6	10	24.50
3	5	8	22.30
4	11	15	21.90
5	13	19	25.60
6	12	22	26.20
7	26	25	27.80
8	11	14	23.80
9	13	18	29.00
10	11	13	27.40

Solution:

The transpose of independent variable matrix X is given by :

$$X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 12 & 10 & 8 & 15 & 19 & 22 & 25 & 14 & 18 & 13 \\ 22.30 & 24.50 & 22.30 & 21.90 & 25.60 & 26.20 & 27.80 & 23.80 & 29.00 & 27.40 \end{bmatrix}^T$$

Similarly, the transpose of dependent variable matrix is given by :

$$Y = [7 \quad 6 \quad 5 \quad 11 \quad 13 \quad 12 \quad 26 \quad 11 \quad 13 \quad 11]^T$$

Now, $(X^T X) = \begin{bmatrix} 10 & 156 & 250.8 \\ 156 & 2692 & 3988.7 \\ 250.8 & 3988.7 & 6347.3 \end{bmatrix}$

Also, $(X^T X)^{-1} = \begin{bmatrix} 13.1157 & 0.1135 & -0.5896 \\ 0.1135 & 0.0064 & -0.0085 \\ -0.5896 & -0.0085 & 0.0288 \end{bmatrix}$

Solution:

Then, $(X^T Y) = \begin{bmatrix} 115 \\ 2041 \\ 2965.7 \end{bmatrix}$

Therefore, $\hat{\beta} = (X^T X)^{-1} X^T Y = \begin{bmatrix} -8.507 \\ 0.882 \\ 0.249 \end{bmatrix}$

Hence, the relationship is given by :

$$E(Y|X_1, X_2) = -8.507 + 0.882X_1 + 0.249X_2$$

Therefore, the MLR model for estimating the system latency of the server $(Y, \text{in ms})$ is :

$E(Y|X_1, X_2) = -8.507 + 0.882X_1 + 0.249X_2$,

where, X_1 is the traffic volume and X_2 is server temperature (in $°C$).

Example – 3

At any given location, the mean annual temperature is estimated from the average elevation (in m) above mean sea level (MSL) and the latitude (°N) by the following model using the data given in Table :

$$E(Y|X_1, X_2) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

where Y: mean annual temperature (in $°C$), X_1: average elevation (in m) above MSL and X_2: latitude(°N).

a) Determine the coefficients β_0, β_1, and β_2 respectively.

b) Using the developed Multiple Linear Regression model, calculate the expected mean annual temperature for the location with an average elevation of 690 m above MSL and a latitude of 34°N.

Example – 3

Observation No.	Mean annual temperature (°C)	Average elevation (m) above MSL	Latitude (°N)
1	25.5	600	33.54
2	30.0	587	26.77
3	26.7	651	31.37
4	28.1	574	29.66
5	29.1	621	27.92
6	26.2	623	32.37
7	26.0	644	32.48
8	25.0	670	33.90
9	24.9	676	33.86
10	28.3	592	29.21

Solution:

a) The independent and dependent variable matrices (X and Y respectively) are given by :

$$X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 600 & 587 & 651 & 574 & 621 & 623 & 644 & 670 & 676 & 592 \\ 33.54 & 26.77 & 31.37 & 29.66 & 27.92 & 32.37 & 32.48 & 33.90 & 33.86 & 29.21 \end{bmatrix}^T$$

$$Y = [25.5 \quad 30.0 \quad 26.7 \quad 28.1 \quad 29.1 \quad 26.2 \quad 26.0 \quad 25.0 \quad 24.9 \quad 28.3]^T$$

Now, $(X^T X) = \begin{bmatrix} 0 & 6200 & 300 \\ 6200 & 3902700 & 194600 \\ 300 & 194600 & 9700 \end{bmatrix}$

Also, $(X^T X)^{-1} = \begin{bmatrix} 34.1884 & -0.0530 & -0.0334 \\ -0.0530 & 0.0002 & -0.0015 \\ -0.0334 & -0.0015 & 0.0302 \end{bmatrix}$

Solution:

Then, $(X^T Y) = \begin{bmatrix} 270 \\ 167890 \\ 8350 \end{bmatrix}$

Therefore, $\hat{\beta} = (X^T X)^{-1} X^T Y = \begin{bmatrix} 49.892 \\ -0.004 \\ -0.656 \end{bmatrix}$

Hence, the relationship is given by :

$$E(Y|X_1, X_2) = 49.892 - 0.004X_1 - 0.656X_2$$

Thus, the MLR model for estimating the mean annual temperature (in $°C$) at a given location is :

$E(Y|X_1, X_2) = 49.9 - 0.004X_1 - 0.656X_2$,

where, X_1 is the elevation (in m) and X_2 is the latitude (in °N).

Solution:

b) The developed MLR model is :

$$E(Y|X_1, X_2) = 49.892 - 0.004X_1 - 0.656X_2$$

Given: X_1 = 690 m and X_2 = 34°N

Substituting these values into the equation,

$$\begin{aligned} Y &= 49.892 - 0.004X_1 - 0.656X_2 \\ &= 49.892 - 0.004 \times 690 - 0.656 \times 34 \\ &= 24.828 \end{aligned}$$

Therefore, the expected mean annual temperature for the location with an average elevation of 690 m above MSL and a latitude of 34°N is 24.8°C.


FREE ONLINE EDUCATION
swayam
Thank You!



SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 06: Simple Linear Regression and Multiple Linear Regression

LECTURE W6L4: Evaluation of Regression Model

BY

Prof. Rajib Maity

Professor, Department of Civil Engineering
Chairperson, School of Water Resources
Indian Institute of Technology Kharagpur
Kharagpur – 721302, West Bengal

CONCEPTS COVERED

Evaluation of Regression Model

NPTEL

Evaluation of Regression Model

- After fitting the regression model over the data, the adequacy of the fitted regression model is required to be checked.
- This can be checked by determining how much of the variability in dependent variable is explained by the regression model.
- The individual value of observed Y, i.e., y_i can be expressed as sum of three components as :

$$y_i = \bar{Y} + (\hat{y}_i - \bar{Y}) + (y_i - \hat{y}_i)$$
$$\Rightarrow (y_i - \bar{Y}) = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{Y})$$

Squaring both sides and summing for all values of Y

$$\sum (y_i - \bar{Y})^2 = \sum (y_i - \hat{y}_i)^2 + \sum (\hat{y}_i - \bar{Y})^2 + 2\sum (y_i - \hat{y}_i)(\hat{y}_i - \bar{Y})$$

Evaluation of Regression Model

- As $\sum (y_i - \hat{y}_i)(\hat{y}_i - \bar{Y}) = 0$ and $\sum (y_i - \bar{Y})^2 = \sum y_i^2 - n\bar{Y}^2$, so the equation can be re-written as:

$$\sum y_i^2 = n\bar{Y}^2 + \sum (y_i - \hat{y}_i)^2 + \sum (\hat{y}_i - \bar{Y})^2$$

- Hence, from the above equation, the total sum of squares of dependent variables $\left(\sum y_i^2\right)$ can be expressed into three following components:
  i. $n\bar{Y}^2$ → sum of squares due to mean (SSM)
  ii. $\sum (y_i - \hat{y}_i)^2$ → sum of squared errors (SSE)
  iii. $\sum (\hat{y}_i - \bar{Y})^2$ → sum of squares due to regression (SSR)
- So, total variability in the dependent variable is the sum of variability explained by regression and variability due to residuals/errors.

Evaluation of Regression Model

- The adequacy of regression model can be expressed as ratio of variability explained by the regression model $\sum(\hat{y}_i - \bar{Y})^2$ and total variability in observed dependent variable $\sum(y_i - \bar{Y})^2$.
- The ratio is called coefficient of determination and represented as r^2 or R^2.

$$R^2 = \frac{\text{Sum of variability in dependent variable explained by regression}}{\text{Total variability in dependent variable}}$$

$$= 1 - \frac{\text{Sum of Squared Error}}{\text{Total variability in dependent variable}}$$

$$= \frac{\sum(\hat{y}_i - \bar{Y})^2}{\sum(y_i - \bar{Y})^2} = \frac{\sum(a + bx_i - a - b\bar{X})^2}{\sum(y_i - \bar{Y})^2} = b^2 \frac{S_{xx}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx}S_{yy}}$$

Evaluation of Regression Model

- The ratio of variability explained by the regression model can never be greater than total variability of the dependent variable.
- Hence, the coefficient of determination ranges between 0 and 1. The closer the R^2 is to 1, the better the model is.
- In the case of multiple linear regression (MLR), coefficient of determination also called coefficient of multiple determination can be calculated using the same equation.
- However, with increase in the number of independent variables, the R^2 will automatically and spuriously increase.
- This may lead to wrong interpretation for the model having large number of independent variables.
- Hence, R^2 needs to be adjusted for increased number of independent variables.

Evaluation of Regression Model

- The adjusted R^2 is always smaller than R^2 and may be negative also.
- Adjusted R^2 (R^2_{adj}) is expressed as :

$$R^2_{adj} = 1 - \frac{\text{Sum of Squared Error}}{\text{Total variability in dependent variable}} \times \frac{n-1}{n-p-1}$$

$$= 1 - (1 - R^2) \times \frac{n-1}{n-p-1}$$

where, n =total no. of observations in dataset and p =no. of independent variables.

- It should be noted that R^2_{adj}, unlike R^2, does not show measure of fit.
- Rather, R^2_{adj} is useful for selecting the variables to be included in an MLR model.
- Apart from R^2, the performance of a prediction/simulation model can be measured using other statistics too.

NPTEL

Evaluation of Regression Model

- Some of other notable performance evaluation measures are:
  - Degree of Agreement (D_r)
  - Nash-Sutcliffe Model Efficiency (NSE)
  - Root Mean Square Error ($RMSE$)
- These performance measure statistics are not limited to linear models only but can be applied to non-linear models too.

NPTEL

Evaluation of Regression Model

Degree of Agreement (D_r):

- The D_r compares the absolute values of deviation between observed and predicted series against the absolute values of deviation of observed values from its mean. It is given by :

$$D_r = \begin{cases} 1 - D_f & for\ D_f \leq 1 \\ \frac{1}{D_f} - 1 & for\ D_f > 1 \end{cases} \qquad \text{where, } D_f = \frac{\sum_{i=1}^{n} |\widehat{Y}_i - Y_i|}{2\sum_{i=1}^{n} |Y_i - \bar{Y}|}$$

The Y_i and $\widehat{Y}_i$ are i^{th} observed and simulated values, respectively, n is length of data and $\bar{Y}$ is mean of observed values.

- The D_r varies between –1 and 1, with higher values showing better model performance.

NPTEL

Evaluation of Regression Model

Nash-Sutcliffe Model Efficiency (NSE):

- NSE measures the model efficiency in terms of relative variance in simulation error compared to variance of observed variable.

$$NSE = 1 - \frac{\sum_i^n (Y_i - \widehat{Y}_i)^2}{\sum_i^n (Y_i - \bar{Y})^2}$$

where, Y_i and $\widehat{Y}_i$ are i^{th} observed and simulated values and $\bar{Y}$ is mean of observed series.

- It should be noted that $\sum_i^n (Y_i - \widehat{Y}_i)^2$ is not necessarily less than $\sum_i^n (Y_i - \bar{Y})^2$.
- Hence, NSE varies in the range of $(-\infty, 1]$, with higher values showing better model efficiency.
- A NSE value of zero means that simulation model is as good as using mean $(\bar{Y})$ as the model output.

NPTEL

Evaluation of Regression Model

Root Mean Square Error ($RMSE$):

- The model performance can also be measured in terms of errors.
- Lower values of prediction error in average sense means better model performance.
- However, as error can be both positive or negative, so performance is measured using mean square sum of error (MSE).
- Positive square root of MSE is known as Root Mean Square Error ($RMSE$).
- $RMSE$ is also known as Root Mean Square Deviation ($RMSD$).

$$MSE = \frac{\sum_i^n \left(\hat{Y}_i - Y_i\right)^2}{n}$$

$$RMSE \text{ or } RMSD = \sqrt{\frac{\sum_i^n \left(\hat{Y}_i - Y_i\right)^2}{n}}$$

NPTEL

Evaluation of Regression Model

- Lower values of MSE and $RMSE$ indicate better model performance.
- One of the criticisms of MSE is that it weighs the outliers heavily as the deviations are quadratically counted in the estimates.
- Same is valid for $RMSE$ too, even though, it is not exactly quadratic.
- This leads to two more variations of $RMSE$: normalized $RMSE$ ($nRMSE$) and unbiased $RMSE$ ($ubRMSE$).
- $nRMSE$ is computed as :

$$nRMSE = \frac{RMSE}{\gamma}$$

where, γ is the normalizing factor, which can be either range $(Y_{max} - Y_{min})$ or mean $(\bar{Y})$ or interquartile range $\left(CDF_Y^{-1}(0.75) - CDF_Y^{-1}(0.25)\right)$ of the observed variable (Y).

NPTEL

Evaluation of Regression Model

- The $ubRMSE$ removes the mean bias $\left(b = mean(\widehat{Y_i}) - mean(Y_i) = \frac{\sum_i^n \widehat{Y_i}}{n} - \frac{\sum_i^n Y_i}{n}\right)$ from the estimate.
- It is computed as :

$$ubRMSE = \sqrt{\frac{\sum_i^n \{[\widehat{Y_i} - mean(\widehat{Y_i})] - [Y_i - mean(Y_i)]\}^2}{n}}$$

- The $ubRMSE$ is related to $RMSE$ through the following equation:

$$RMSE^2 = ubRMSE^2 + b^2$$

NPTEL

Example – 1

Monthly precipitation $(X, \text{in cm})$ and runoff $(Y, \text{in cm})$ for a location were recorded over a period of two years and are presented in Table on the next slide. The variables are assumed to be linearly related.

a) Evaluate the coefficient of determination between observed (Y) and model values $(\hat{Y})$ of runoff obtained through a linear regression model.

NPTEL

Example – 1

Sl. No. (Month)	X(cm)	Y(cm)	Sl. No. (Month)	X(cm)	Y(cm)
1	6.9	2.40	13	5.5	1.40
2	6.4	1.10	14	11.4	6.30
3	6.5	1.70	15	10.8	4.40
4	5.1	0.50	16	7.5	1.80
5	7.1	1.80	17	8.2	4.20
6	7.1	2.00	18	7.9	2.90
7	10.2	4.20	19	4.1	0.00
8	9.9	3.00	20	5.0	1.30
9	8.4	1.70	21	6.7	1.30
10	5.8	3.40	22	4.3	0.00
11	10.1	4.40	23	10.4	5.90
12	7.3	2.80	24	3.9	2.40

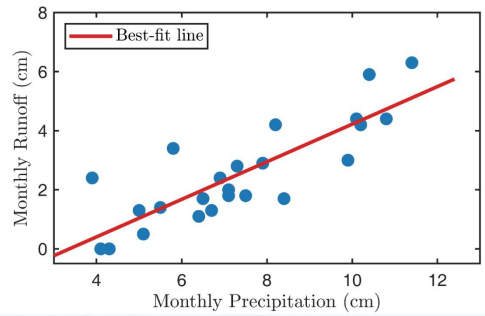
Solution:

Given:

Independent Variable, X = Precipitation (in cm)

Total no. of months, $n = 24$

Dependent Variable, Y = Runoff (in cm)



The relationship between X and expected value of Y, being linear, is expressed as:

$$\hat{Y} = bX + a$$

The calculations required to estimate the parameters (a and b) are presented in the table (in the next slide).

Solution:

S. No	y_i	x_i	x_i^2	$x_i y_i$	y_i^2	S. No	y_i	x_i	x_i^2	$x_i y_i$	y_i^2
1	2.40	6.90	47.61	16.56	5.76	13	1.40	5.50	30.25	7.70	1.96
2	1.10	6.40	40.96	7.04	1.21	14	6.30	11.40	129.96	71.82	39.69
3	1.70	6.50	42.25	11.05	2.89	15	4.40	10.80	116.64	47.52	19.36
4	0.50	5.10	26.01	2.55	0.25	16	1.80	7.50	56.25	13.50	3.24
5	1.80	7.10	50.41	12.78	3.24	17	4.20	8.20	67.24	34.44	17.64
6	2.00	7.10	50.41	14.20	4.00	18	2.90	7.90	62.41	22.91	8.41
7	4.20	10.20	104.04	42.84	17.64	19	0.00	4.10	16.81	0.00	0.00
8	3.00	9.90	98.01	29.70	9.00	20	1.30	5.00	25.00	6.50	1.69
9	1.70	8.40	70.56	14.28	2.89	21	1.30	6.70	44.89	8.71	1.69
10	3.40	5.80	33.64	19.72	11.56	22	0.00	4.30	18.49	0.00	0.00
11	4.40	10.10	102.01	44.44	19.36	23	5.90	10.40	108.16	61.36	34.81
12	2.80	7.30	53.29	20.44	7.84	24	2.40	3.90	15.21	9.36	5.76

Totals	Values
$\sum_{i=1}^{n} y_i$	60.90
$\sum_{i=1}^{n} x_i$	176.50
$\sum_{i=1}^{n} x_i^2$	1410.51
$\sum_{i=1}^{n} x_i y_i$	519.42
$\sum_{i=1}^{n} y_i^2$	219.89

Solution:

The parameter b can be calculated as:

$$S_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n} = 519.42 - \frac{(176.50 \times 60.90)}{24} = 71.55$$

$$S_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} = 1410.51 - \frac{(176.50)^2}{24} = 112.50$$

$$b = \frac{S_{xy}}{S_{xx}} = \frac{71.55}{112.50} = 0.636$$

The parameter a is calculated as:

$$a = \frac{1}{n}\left(\sum_{i=1}^{n} y_i - b \sum_{i=1}^{n} x_i\right) = \frac{1}{24}(60.90 - 0.636 \times 176.50) = -2.14$$

NPTEL

Solution:

Hence, the regression equation is : $\hat{Y} = 0.636X - 2.14$

Also, $S_{yy} = \sum_{i=1}^{n} y_i^2 - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n} = 219.89 - \frac{(60.90)^2}{24} = 65.36$

Hence, the coefficient of determination is given by:

$$R^2 = b^2 \frac{S_{xx}}{S_{yy}} = 0.636^2 \times \frac{112.50}{65.36} = 0.696 \approx 0.70$$

Example – 2

A data scientist aims to model the linear relationship between product defects and assembly line's operating conditions. Over ten distinct observation periods, three variables were recorded (shown in the table): Defect Count, Conveyor Speed (the speed of the assembly line, m/min) and Ambient Temperature (the factory floor temperature near the heavy machinery, measured in $°C$).

It is assumed that expected defect count (Y) can be estimated from conveyor speed (X_1) and ambient temperature (X_2) using MLR model given by:

$$E(Y|X_1, X_2) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

Determine the coefficients β_0, β_1, and β_2. Also evaluate R^2, R^2_{adj}, NSE, $RMSE$ and D_r for the developed model.

Example – 2

Observation No.	Defect Count (Y)	Conveyor Speed (X_1, m/min)	Ambient Temperature (X_2, °C)
1	7	12	22.30
2	6	10	24.50
3	5	8	22.30
4	11	15	21.90
5	13	19	25.60
6	12	22	26.20
7	26	25	27.80
8	11	14	23.80
9	13	18	29.00
10	11	13	27.40

Solution:

The transpose of independent variable matrix X is given by:

$$X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 12 & 10 & 8 & 15 & 19 & 22 & 25 & 14 & 18 & 13 \\ 22.30 & 24.50 & 22.30 & 21.90 & 25.60 & 26.20 & 27.80 & 23.80 & 29.00 & 27.40 \end{bmatrix}^T$$

Similarly, the transpose of dependent variable matrix Y is given by:

$$Y = [7 \quad 6 \quad 5 \quad 11 \quad 13 \quad 12 \quad 26 \quad 11 \quad 13 \quad 11]^T$$

Now, $(X^T X) = \begin{bmatrix} 10 & 156 & 250.8 \\ 156 & 2692 & 3988.7 \\ 250.8 & 3988.7 & 6347.3 \end{bmatrix}$

Also, $(X^T X)^{-1} = \begin{bmatrix} 13.1157 & 0.1135 & -0.5896 \\ 0.1135 & 0.0064 & -0.0085 \\ -0.5896 & -0.0085 & 0.0288 \end{bmatrix}$

Solution:

Then, $(X^T Y) = \begin{bmatrix} 115 \\ 2041 \\ 2965.7 \end{bmatrix}$

Therefore, $\hat{\beta} = (X^T X)^{-1} X^T Y = \begin{bmatrix} -8.507 \\ 0.882 \\ 0.249 \end{bmatrix}$

Hence, the relationship is given by :

$$E(Y|X_1, X_2) = -8.507 + 0.882X_1 + 0.249X_2$$

Parameters required to determine R^2 and R^2_{adj}, are evaluated and tabulated in the table (in next slide), where,

$\hat{Y} = -8.507 + 0.882X_1 + 0.249X_2$ and

$\bar{Y} = \frac{\sum Y}{n} = \frac{7+6+5+11+13+12}{10} = \frac{115}{10} = 11.50$

Solution:

Obs No.	Y	X_1	X_2	$\hat{Y}$	$\lvert Y - \hat{Y} \rvert$	$(Y - \hat{Y})^2$	$\lvert Y - \bar{Y} \rvert$	$(Y - \bar{Y})^2$
1	7	12	22.30	7.63	-0.63	0.40	-4.50	20.25
2	6	10	24.50	6.41	-0.41	0.17	-5.50	30.25
3	5	8	22.30	4.10	0.90	0.81	-6.50	42.25
4	11	15	21.90	10.18	0.82	0.68	-0.50	0.25
5	13	19	25.60	14.63	-1.63	2.64	1.50	2.25
6	12	22	26.20	17.42	-5.42	29.39	0.50	0.25
7	26	25	27.80	20.47	5.53	30.63	14.50	210.25
8	11	14	23.80	9.77	1.23	1.52	-0.50	0.25
9	13	18	29.00	14.59	-1.59	2.53	1.50	2.25
10	11	13	27.40	9.78	1.22	1.48	-0.50	0.25

Totals	Values
$\sum_{i=1}^{n} \lvert Y - \hat{Y} \rvert$	19.39
$\sum_{i=1}^{n} \lvert Y - \bar{Y} \rvert$	36.00
$\sum_{i=1}^{n} (Y - \hat{Y})^2$	70.25
$\sum_{i=1}^{n} (Y - \bar{Y})^2$	308.50

Solution:

Sum of squared errors (SSE) = $\sum_{i=1}^{n}\left(Y-\hat{Y}\right)^2 = \sum_{i=1}^{n}(Y-X\beta)^2 = 70.25$

Total Variance in dependent variable = $\sum_{i=1}^{n}(Y-\bar{Y})^2 = 308.50$

Coefficient of determination, $R^2 = 1-\frac{\sum_{i=1}^{n}(Y-\hat{Y})^2}{\sum_{i=1}^{n}(Y-\bar{Y})^2} = 1-\frac{70.25}{308.50} = 0.77$

Adjusted Coefficient of determination,

$R^2_{adj} = 1-(1-R^2)\times\frac{n-1}{n-p-1} = 1-(1-0.77)\times\frac{10-}{10-2-1} = 0.70$

Therefore, coefficient of determination and adjusted coefficient of determination of the developed MLR model are 0.77 and 0.70 respectively.

Solution:

Now, $NSE = 1-\frac{\sum_i^n(Y_i-\widehat{Y_i})^2}{\sum_i^n(Y_i-\bar{Y})^2}$

$$NSE = 1-\frac{70.25}{308.50} = 1-0.2277 = 0.7723$$

Also, $RMSE = \sqrt{\frac{\sum_i^n(\widehat{Y_i}-Y_i)^2}{n}} = \sqrt{\frac{70.25}{10}} = \sqrt{7.025} = 2.65$

And, $D_r = \begin{cases} 1-D_f & for\ D_f \le 1 \\ \frac{1}{D_f}-1 & for\ D_f > 1 \end{cases}$ where, $D_f = \frac{\sum_{i=1}^{n}|\widehat{Y_i}-Y_i|}{2\sum_{i=1}^{n}|Y_i-\bar{Y}|} = \frac{19.39}{2\times36} = 0.2693$

As, $D_f \le 1$, so, $D_r = 1-D_f = 1-0.2693 = 0.73$

Therefore, NSE, $RMSE$ and D_r values for the developed MLR model are: 0.77, 2.65 units and 0.73 respectively.


FREE ONLINE EDUCATION
swayam
Thank You!
alt



SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 06: Simple Linear Regression and Multiple Linear Regression

LECTURE W6L5: Uncertainty Quantification of Regression Model

BY

Prof. Rajib Maity

Professor, Department of Civil Engineering
Chairperson, School of Water Resources
Indian Institute of Technology Kharagpur
Kharagpur – 721302, West Bengal

CONCEPTS COVERED

- Correlation and Regression
- Correlation and Causality
- Uncertainty Quantification of Regression Model
  - Basics of Hypothesis Testing
- Discussion with Examples

Correlation and Regression

- Coefficient of correlation is a measure of linear association between dependent and independent variables.
- Mathematically, sample correlation coefficient is defined as the sum of product of standardized variable divided by $(n - 1)$.

$$r = \frac{1}{n-1}\sum\left(\frac{x_i-\bar{X}}{\sqrt{S_{xx}}}\right)\left(\frac{y_i-\bar{Y}}{\sqrt{S_{yy}}}\right) = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{\sqrt{S_{xx}}S_{xy}}{\sqrt{S_{yy}}S_{xx}} = \frac{\sqrt{S_{xx}}}{\sqrt{S_{yy}}}b$$

where b is least squares estimate of slope for simple linear regression model (β).

- If most of the deviation from mean in either of X or Y are of same sign then, the r will be having positive value.
- In other words, if both the variables deviate from mean in similar trend (one increases then other also increases and vice versa) then the linear association is high.

Correlation and Regression

- The correlation coefficient can also become negative if most of the deviation from mean in either of X or Y have opposite signs.
- The magnitude of r ranges between –1 and 1.
- Based on the value of correlation coefficient, following inferences about the linear relationship between the variables can be drawn:

i. The magnitude and sign of r represent the strength of linear association and direction of slope of the fitted straight line between variables, respectively.

ii. A value of r closer to zero represents very weak linear association between the variables involved. In such cases, simple linear regression may not be able to model the relationship between the variables.

Example –1

For a dataset containing two variables X and Y, a simple linear regression model was fitted, and the following information was obtained:

Regression equation: $\hat{Y} = 0.636X - 2.14$

Sum of squares for X: $S_{xx} = 112.50$

Sum of squares for Y: $S_{yy} = 65.36$

Using the given information, determine the coefficient of correlation (r) between the variables X and Y.

Solution:

For a simple linear regression model, $\hat{Y} = 0.636X - 2.14$, the coefficient of correlation r is related to the regression slope b and the sums of squares by:

$$r = \frac{\sqrt{S_{xx}}}{\sqrt{S_{yy}}} b$$

In this problem,

$b = 0.636$, $S_{xx} = 112.50$ and $S_{yy} = 65.36$

Therefore, coefficient of correlation between (X) and (Y) is given by:

$$r = \frac{\sqrt{S_{xx}}}{\sqrt{S_{yy}}} b = \frac{\sqrt{112.50}}{\sqrt{65.36}} \times 0.636 = 0.834$$

Hence, the coefficient of correlation between X and Y is 0.834, which indicates a strong positive linear relationship between the two variables. As X increases, Y tends to increase as well.

Correlation and Causality

- A high observed correlation does not suggest anything about a cause-and-effect relationship.
- Hence, the observation that two variables tend to vary simultaneously in the same direction does not imply a direct causal relationship between them.
- Both variables may depend on unknown variables and positive correlation is being produced due to the mutual relationship with other variables.
- These unknown variables are called lurking variables.
- Lurking variables are often overlooked when mistaken claims are made about X causing Y.
- Hence, the correlation coefficient should not be taken as a measure of relationship or causality.
- Sometimes, a causal relationship may also exist that is opposite with respect to the sign of the observed correlation.

Uncertainty Quantification of Regression Model: Hypothesis Testing

- Hypothesis testing is the process of accepting or rejecting an assumption regarding a population parameter, which may or may not be true.
- There are many problems in which we must decide whether a statement concerning a parameter is true or false, that is, we must test the hypothesis about a parameter.
- Procedures that enable us to reject or fail to reject a hypothesis or to determine whether observed sample statistics differ significantly from population parameters are called hypothesis tests.
- It is always useful to make assumptions about the population involved.
- Initially, it is needed to decide the neutral or by default assumptions. This is denoted as null hypothesis (H_0).
- The opposite of null hypothesis (H_0), that we want to test is assigned as alternative hypothesis (H_a).

Type I and Type II Errors

True fact	Decision-making	
	Accept	Reject
Hypothesis H is true	Correct	Type I error
Hypothesis H is false	Type II error	Correct

- The probability of committing Type I error when the hypothesis is true is designated by α also known as level of significance.
- The probability of committing Type II error when the hypothesis is false is designated by β.
- Our major aim is to minimize the error which is generally achieved by fixing the value of α and minimizing β as far as possible.

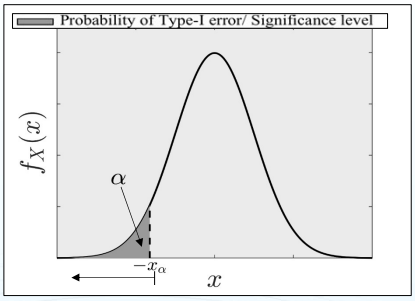
Tests of Hypothesis

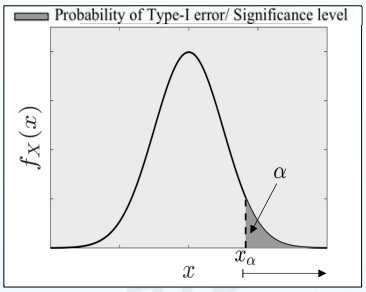
To approach the problem of hypothesis testing systematically, it will help to proceed as outlined in the following steps:

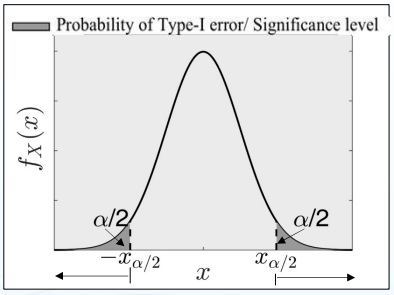
i. Formulation of a null hypothesis and an appropriate alternative hypothesis, which can be considered to hold as true when the null hypothesis is rejected.
ii. Specification of the probability of Type I error or significance level, designated by α.
iii. Based on the sampling distribution of an appropriate statistic, a criterion for testing the null hypothesis against the alternative is constructed.
iv. From the data, the value of the test statistic on which the decision is to be based is evaluated.
v. The final decision is whether to ***reject*** the null hypothesis or whether to ***fail to reject*** the null hypothesis.

The null hypothesis is rejected only if the test statistic falls within the critical zone defined by the significance level.

Critical Zone: One sided and Two-sided Test



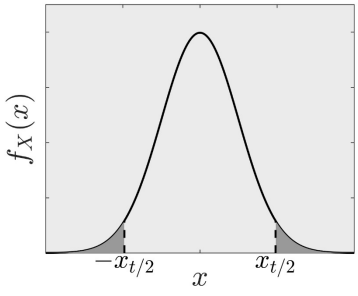


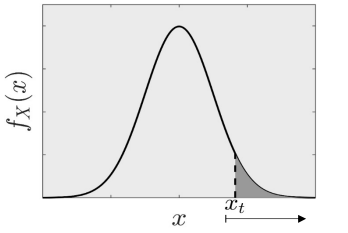


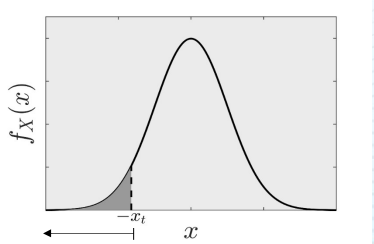
- Considering α as the significance level, the critical zone or rejection zone for one-sided test is either $[lb, x_{1-\alpha}]$ or $[x_\alpha, ub]$ based on the hypothesis to be tested.
- For the two-sided test, the critical zones are $[lb, x_{1-\alpha/2}]$ and $[x_{\alpha/2}, ub]$.
- In all the cases (i.e., one-sided and two-sided), the null hypothesis is rejected if the test statistic lies in the critical zone.

p-value

- The p-value is the probability of obtaining a value of the test statistic that is as extreme as, or more extreme than, the value actually observed.
- The concept of p-value with respect to the test statistic (x_t) for one-sided or two-sided tests is illustrated in the figures.
- Area under the curve is equal to the p-value.







Summary of Hypothesis Tests

Test Type	Case	Parameter Tested
Single Sample Test	One mean when the population variance is known	Population mean, μ
	One mean when the population variance is unknown	**Population mean, μ**
	One variance	Population variance, σ^2
Two Sample Test	Two means when the population variances are known	Difference between two means, $\mu_1 - \mu_2$
	Two means when the population variances are unknown	Difference between two means, $\mu_1 - \mu_2$
	Two variances	Ratio of Variances $\left(\frac{\sigma_1^2}{\sigma_2^2}\right)$
Proportion Test	One proportion (assuming a sufficiently large sample size)	Population proportion, p
	Multiple proportions (testing if $p_1 = p_2 = \ldots = p_k$)	Several proportions, p_1, p_2, …, p_k

Single Sample Test

Hypothesis concerning one mean when the variance is unknown

- In this case, the variance of the population is unknown and calculated from the sample and the test statistic is as follows:

$$\text{Test Statistic : } t = \frac{\bar{x} - \mu_0}{\frac{S}{\sqrt{n}}}$$

where $\bar{x}$ is the sample mean, S is the sample variance and μ_0 is a particular value of mean for which the hypothesis is to be tested.

- This test statistic follows student's t distribution with degrees of freedom $(n-1)$.

Hypothesis concerning one mean when the variance is unknown

The rejection criterion for three different cases is shown as follows:

H_0	H_a	Rejection region
$\mu = \mu_0$	$\mu \neq \mu_0$	$\lvert t \rvert > t_{\alpha/2}\,(n-1)$
$\mu \geq \mu_0$	$\mu < \mu_0$	$t < -t_{\alpha}\,(n-1)$
$\mu \leq \mu_0$	$\mu > \mu_0$	$t > t_{\alpha}\,(n-1)$

Uncertainty Quantification of Regression Parameters

- Uncertainty quantification of regression parameters can be expressed through confidence interval.
- The confidence interval of least squares estimates of α and β (i.e., the least squares estimators of the regression coefficients) depends on the estimate of standard error.
- The standard error σ^2 is estimated from the deviation of sample points from the estimated least squares line. The estimate of σ^2 from a sample is given by standard error of estimate (S_e).
- Standard error of estimate is the residual sum of squares or the sum of squared errors divided by $(n-2)$ and is expressed as :

$$S_e^2 = \frac{1}{n-2}\sum(y_i - \hat{y}_i)^2 = \frac{S_{yy} - (S_{xy})^2/S_{xx}}{n-2}$$

Confidence Interval

- Following two statistics are defined as follows for α and β, respectively:

$$t^a = \frac{a-\alpha}{S_e}\sqrt{\frac{nS_{xx}}{S_{xx}+n(\bar{x})^2}} \quad \text{and} \quad t^b = \frac{(b-\beta)}{S_e}\sqrt{S_{xx}}$$

- These statistics t^a and t^b follow t-distribution with $n-2$ degrees of freedom.
- To construct confidence intervals of $(1-\alpha)100\%$ for the regression coefficients α and β, the appropriate t-statistic is substituted into $-t_{\alpha/2} < t < t_{\alpha/2}$.

- This leads to the following confidence intervals:

$$-t_{\alpha/2} < \frac{a-\alpha}{S_e}\sqrt{\frac{nS_{xx}}{S_{xx}+n(\bar{x})^2}} < t_{\alpha/2} \quad \text{and} \quad -t_{\alpha/2} < \frac{(b-\beta)}{S_e}\sqrt{S_{xx}} < t_{\alpha/2}$$

Confidence Interval

- After re-arrangement, the following confidence intervals are expressed as:

$$a - t_{\alpha/2}S_e\sqrt{\frac{1}{n}+\frac{(\bar{x})^2}{S_{xx}}} < \alpha < a + t_{\alpha/2}S_e\sqrt{\frac{1}{n}+\frac{(\bar{x})^2}{S_{xx}}}$$

$$b - t_{\alpha/2}S_e\frac{1}{\sqrt{S_{xx}}} < \beta < b + t_{\alpha/2}S_e\frac{1}{\sqrt{S_{xx}}}$$

Confidence Interval

- The estimate $\hat{y}\ (= a + bx)$ follows a t-distribution with mean $a + bx$, variance $S_e{}^2\left(\frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}\right)$ and degrees of freedom $(n-2)$.
- Thus, $(1-\alpha)100\%$ confidence interval of the estimated value $(\hat{y})$ is given by :

$$\hat{y}: (a + bx) \pm t_{\alpha/2} S_e \sqrt{\frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}}$$

- After re-arrangement, the following confidence intervals are expressed as

$$(a + bx) - t_{\alpha/2} S_e \sqrt{\frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}} < \hat{y} < (a + bx) + t_{\alpha/2} S_e \sqrt{\frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}}$$

Example –2

An agricultural researcher is developing a model to predict crop yield (Y, in tons/hectare) based on fertilizer application (X, in kg/hectare). A Simple Linear Regression (SLR) model was fitted to 24 field plots ($n = 24$), yielding the following estimated coefficients and summary statistics:

- Intercept, $a = -2.220$
- Mean of X, $\bar{x} = 7.354$
- Slope, $b = 0.664$
- Sum of squares for X,
  $S_{xx} = 112.50$
- Sum of squared errors variance,
  $S_e^2 = 0.88$

Example –2

Using the given information, evaluate:

a) Determine the 95% confidence interval for the intercept parameter (α).

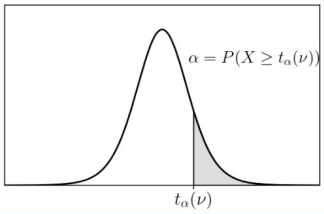
b) Is the slope parameter (β) is equal to unity at 5% level of significance.

c) Check whether the regression line passes through the origin at 0.01 level of significance.

NPTEL

Student's t – distribution percentage points

	α								
ν	0.25	0.1	0.05	0.025	0.01	0.00833	0.00625	0.005	0.0025
1	1.000	3.078	6.314	12.706	31.821	38.204	50.923	63.657	127.321
2	0.816	1.886	2.920	4.303	6.965	7.650	8.860	9.925	14.089
3	0.765	1.638	2.353	3.182	4.541	4.857	5.392	5.841	7.453
4	0.741	1.533	2.132	2.776	3.747	3.961	4.315	4.604	5.598
5	0.727	1.476	2.015	2.571	3.365	3.534	3.810	4.032	4.773
6	0.718	1.440	1.943	2.447	3.143	3.288	3.521	3.707	4.317
7	0.711	1.415	1.895	2.365	2.998	3.128	3.335	3.499	4.029
8	0.706	1.397	1.860	2.306	2.896	3.016	3.206	3.355	3.833
9	0.703	1.383	1.833	2.262	2.821	2.934	3.111	3.250	3.690
10	0.700	1.372	1.812	2.228	2.764	2.870	3.038	3.169	3.581
11	0.697	1.363	1.796	2.201	2.718	2.820	2.981	3.106	3.497
12	0.695	1.356	1.782	2.179	2.681	2.780	2.934	3.055	3.428
13	0.694	1.350	1.771	2.160	2.650	2.746	2.896	3.012	3.372
14	0.692	1.345	1.761	2.145	2.624	2.718	2.864	2.977	3.326
15	0.691	1.341	1.753	2.131	2.602	2.694	2.837	2.947	3.286
16	0.690	1.337	1.746	2.120	2.583	2.673	2.813	2.921	3.252
17	0.689	1.333	1.740	2.110	2.567	2.655	2.793	2.898	3.222
18	0.688	1.330	1.734	2.101	2.552	2.639	2.775	2.878	3.197
19	0.688	1.328	1.729	2.093	2.539	2.625	2.759	2.861	3.174
20	0.687	1.325	1.725	2.086	2.528	2.613	2.744	2.845	3.153
21	0.686	1.323	1.721	2.080	2.518	2.602	2.732	2.831	3.135
22	0.686	1.321	1.717	2.074	2.508	2.591	2.720	2.819	3.119
23	0.685	1.319	1.714	2.069	2.500	2.582	2.710	2.807	3.104
24	0.685	1.318	1.711	2.064	2.492	2.574	2.700	2.797	3.091
25	0.684	1.316	1.708	2.060	2.485	2.566	2.692	2.787	3.078
26	0.684	1.315	1.706	2.056	2.479	2.559	2.684	2.779	3.067
27	0.684	1.314	1.703	2.052	2.473	2.553	2.676	2.771	3.057
28	0.683	1.313	1.701	2.048	2.467	2.547	2.669	2.763	3.047
29	0.683	1.311	1.699	2.045	2.462	2.541	2.663	2.756	3.038
30	0.683	1.310	1.697	2.042	2.457	2.536	2.657	2.750	3.030
40	0.681	1.303	1.684	2.021	2.423	2.499	2.616	2.704	2.971
50	0.679	1.299	1.676	2.009	2.403	2.477	2.591	2.678	2.937
60	0.679	1.296	1.671	2.000	2.390	2.463	2.575	2.660	2.915
120	0.677	1.289	1.658	1.980	2.358	2.428	2.536	2.617	2.860
1000	0.675	1.282	1.646	1.962	2.330	2.398	2.502	2.581	2.813



Solution:

Given:

Independent Variable, X = Fertilizer application (in kg/hectare)

Total no. of runs, $n = 24$

$\Rightarrow n - 2 = 24 - 2 = 22$

Dependent Variable, Y = Crop yield (in tons/hectare)

Intercept $(a) = -2.220$

Mean of $X(\bar{x}) = 7.354$

Slope $(b) = 0.664$

Sum of squares for $X(S_{xx}) = 112.50$

Sum of squared errors variance $(S_e^2) = 0.88$

$$\Rightarrow S_e = \sqrt{0.88} = 0.938$$

NPTEL

Solution:

a) 95% Confidence Interval for Intercept (α)

For $(n-2) = 22$ degrees of freedom, $t_{0.975} = 2.074$

Thus, 95% confidence limit for parameter α is given by:

$$\alpha: a \pm t_{\alpha/2} S_e \sqrt{\frac{1}{n} + \frac{(\bar{x})^2}{S_{xx}}} = (-2.22) \pm 2.074 \times 0.938 \sqrt{\frac{1}{24} + \frac{(7.354)^2}{112.5}}$$

Hence, the confidence interval for intercept parameter α is given by:

$$-3.626 \leq \alpha \leq -0.813$$

b) The hypothesis that the slope parameter (β) is equal to unity at 5% level of significance

The null and alternate hypothesis can be expressed as :

Null Hypothesis: $\beta = 1$.

Solution:

Alternative Hypothesis: $\beta \neq 1$.

Level of significance $= 0.05$.

For $(n-2) = 22$ degrees of freedom,

$$t_{0.975} = 2.074 \text{ and } t_{0.025} = -2.074$$

Hence, the critical zone is given by $(-\infty, -2.074]$ *and* $[2.074, \infty)$.

The test statistic is given by:

$$t^b = \frac{(b-\beta)}{S_e}\sqrt{S_{xx}} = \frac{(0.664-1)}{0.938}\sqrt{112.5} = -3.799$$

- Since t^b falls in the critical rejection zone (as it is less than -2.074), the null hypothesis must be rejected.
- Hence, the slope parameter β is statistically not equal to 1.

Solution:

c) Checking if the regression line passes through origin

If the regression line passes through origin, then its intercept on the y-axis should be 0. So, α should be zero. Hence,

Null Hypothesis: $\alpha = 0$.

Alternative Hypothesis: $\alpha \neq 0$.

Level of significance $= 0.01$.

For $(n-2) = 22$ degrees of freedom,

$$t_{0.995} = 2.819 \text{ and } t_{0.005} = -2.819$$

Hence, the critical zone is given by

$(-\infty, -2.819]$ *and* $[2.819, \infty)$

Solution:

The test statistic is given by :

$$t^a = \frac{a-\alpha}{S_e}\sqrt{\frac{nS_{xx}}{S_{xx}+n(\bar{x})^2}} = \frac{-2.22-0}{0.938}\sqrt{\frac{24\times112.5}{112.5+24\times(7.35)^2}} = -3.273$$

- Since t^a falls in the critical rejection zone (as it is less than -2.819), the null hypothesis is rejected at 0.01 significance level.
- Hence, the regression line statistically does not pass through the origin.

